

# Heirloom Recipe Collection & AI Studio

## Comprehensive Application Feature Specification & Architecture Guide

**Overview:** The Heirloom Recipe Application is a state-of-the-art, local-first culinary archive and kitchen companion engineered for Android (Samsung Galaxy Tab S8+ and Galaxy Phone). It synthesizes multimodal Vision AI (Google Gemini 2.5 Flash), local GPU image synthesis (ComfyUI with official Z-Image Turbo and SDXL Turbo models), multi-profile cookbook separation, hands-free cooking mode, and self-hosted multi-device synchronization.

### 1. Multimodal Recipe Scanning & Vision AI Pipeline

- **Continuous Rapid Camera Loop**

Auto-reopens the camera between pages (Page 1 -> Page 2 -> Page 3+) and immediately processes and saves upon completion without intermediate review interruptions.

- **Physical Card Snapshot Preservation**

Captured camera snapshots and gallery photos are saved directly to local storage and preserved in the recipe booklet so heirloom cards are never lost.

- **Bilingual & Cursive Transcription**

Transcribes historical German handwritten script (Kurrentschrift, Sütterlin) and modern cursive, creating natural English translations while retaining original German text.

- **Strict Ingredient Extraction Fidelity**

Captures all leavening agents, extracts, fractions (e.g. 2 1/4, 1/2), units, sub-groups (Dough, Filling, Glaze), and package descriptions without omission.

- **Automatic Food Photo Cropping**

Detects food photographs embedded within physical recipe cards and automatically crops them as recipe cover photos.

## 2. AI Photo Generation Engine & Cover Art Studio

- **Official Z-Image Turbo Architecture**

Direct API execution via local PC ComfyUI (<http://192.168.1.54:8188>) using UNETLoader (z\_image\_turbo\_bf16), Qwen 3.4B text encoder (lumina2), and ModelSamplingAuraFlow (shift 3.0).

- **SDXL Turbo Checkpoint Integration**

Native 1-second generation using sd\_xl\_turbo\_1.0\_fp16.safetensors via standard CheckpointLoaderSimple.

- **Culinary Resolution Optimization (768 x 768)**

Optimized latent grid delivering sharp food textures, realistic lighting, and fast execution without memory bloating.

- **Img2Img Photo Synthesis**

Enhances physical card snapshots into photorealistic dish presentations while retaining the original plating geometry.

- **Google Gemini Cloud Fallback**

Alternative cloud-based image generation engine powered by Google AI Studio.

- **Batch AI Cover Studio**

Scans the entire cookbook for recipes without photos and generates covers automatically in the background.

## 3. Interactive Kitchen Sous-Chef Assistant

- **Live Conversational AI**

Powered by Google Gemini 2.5 Flash with live streaming response generation.

- **Voice & Natural Language Intent Parsing**

Understands cooking questions, substitutions, recipe lookups, and app actions.

- **Automated Action Execution**

Triggers scanning ('Scan recipes for Annette'), URL import ('Extract recipe from link'), profile switching, and bulk recipe moving directly from chat.

## 4. Web Recipe Importer (URL Extraction)

- **Universal URL Scraping**

Fetches and parses recipes from food blogs, cooking sites, and YouTube descriptions.

- **Structured Clean Extraction**

Strips ads, popups, and filler stories, extracting ingredients, steps, timings, yields, and high-res cover photos.

## 5. 3D Skeuomorphic Recipe Booklet Experience

- **Realistic 3D Page Curl & Turn Animations**

Smooth 3D page-turning physics with camera perspective, shadow depth, and page sound effects.

- **Interactive Multi-Page Structure**

Page 0: Embossed Hardcover & Food Photo • Page 1: Lore, Origin Story & TOC • Page 2: Scalable Ingredients • Page 3+: Step-by-Step Cooking • Last: Cook's Journal.

- **Tablet Two-Page Open Book Spread**

Wide-screen split layout showing Left & Right pages simultaneously with stitched spine depth on Galaxy Tab S8+.

- **Vintage Handwritten Card View**

Split view displaying the original physical card scan side-by-side with digital instructions.

## 6. Smart Scaler, Unit & German Culinary Converter

- **Dynamic Serving Scaler**

Instant fractional and decimal ingredient recalculation (0.5x, 1x, 1.5x, 2x, 3x, 4x).

- **German Culinary Unit Converter**

Converts traditional German volume measures (Esslöffel, Teelöffel, Prise, Messerspitze, Tasse) to metric grams or US ounces.

- **Ingredient Density Calculations**

Calculates true weight-to-volume ratios based on ingredient type (flour, sugar, butter, liquids).

- **Bilingual Culinary Glossary**

Built-in dictionary explaining traditional German cooking terms, ingredients, and techniques.

## 7. Hands-Free Kitchen Cook Mode

- **High-Visibility Distraction-Free Layout**

Large typography and high-contrast step cards designed for across-the-counter reading.

- **Multi-Step Countdown Timers**

Automatic timer extraction from recipe steps with audio alarm alerts upon completion.

- **Text-to-Speech (TTS) Voice Narration**

Reads steps aloud in English or German for hands-free cooking.

- **Keep Screen On Lock**

Prevents device display timeout during active cooking sessions.

## 8. Smart Categorized Shopping List

- **1-Tap Ingredient Export**

Adds scaled recipe ingredients directly to the grocery list.

- **Supermarket Aisle Categorization**

Automatically organizes items into Produce, Dairy, Pantry, Meat, Spices, and Bakery.

- **Badge Counter & Progress Tracking**

Interactive checkbox list with uncompleted item badge counters on navigation bars.

## 9. Multi-Profile & Family Cookbook Management

- **Independent Family Profiles**

Separate collections for family members (e.g. 'Louis', 'Annette', 'All Family').

- **1-Tap Bulk Recipe Move**

Bulk moves recipes between family profiles with automatic multi-device synchronization.

- **Custom Category Hierarchy**

Create, rename, and delete custom recipe categories (Baking, Main Dishes, Soups, etc.).

## 10. Cloud Hub & Multi-Device Synchronization

- **Self-Hosted FastAPI / SQLite Backend**

Bidirectional delta synchronization with your private cloud hub.

- **Automatic Multi-Device Mirroring**

Seamlessly syncs recipes and cover photos across Tablet and Phone.

- **Wi-Fi Auto-Sync & Background Polling**

Silently synchronizes changes when connected to home Wi-Fi or when opening the app.

## 11. Data Safety, Backups & Printable PDF Export

- **Automatic Weekly Backups**

Scheduled local SQLite backups protecting your culinary heritage.

- **Full ZIP Export & Restore**

Export complete archive with images and metadata for offline safekeeping.

- **Printable PDF Recipe Cards**

Generates high-resolution formatted PDF recipe cards for physical binders and sharing.